

# China Plus X: A New Logistics Map of Asia

## Preview

---

Following China Plus One, Asia's production landscape is expanding beyond China to Vietnam, Malaysia, Thailand, Indonesia, and India, rather than moving away from China altogether. China is broadening its role as a vast manufacturing "platform" that supplies components, intermediate goods, and equipment. As more components move across borders, new demand for air cargo is emerging.

This white paper defines this phenomenon as "China Plus X." It examines air cargo growth in the Asia-Pacific region using statistics from IATA and ACI, and analyzes how the changing industrial roles of five countries are affecting competition among Asia's air cargo hubs. It demonstrates that China Plus X presents an opportunity, rather than a threat, for Korean air logistics companies, and suggests that a multi-hub network strategy centered on Incheon International Airport, together with supply chain visibility capabilities, will become a key competitive advantage.





About the Author

## Professor Namjin Baek

Professor Namjin Baek is a leading researcher in the field of ICT- and drone-based air logistics systems.

Drawing on his professional experience in the aviation industry, his research expertise spans air logistics systems, warehousing and cargo handling, aviation management, air transportation services, warehouse management systems (WMS), transportation management systems (TMS), radio-frequency identification (RFID), unmanned aerial vehicles (UAVs), and urban air mobility (UAM). He is committed to advancing the aviation industry by integrating data analytics, programming, and computer networking technologies.

He also serves as a technical review and advisory committee member and evaluator for projects involving major organizations, including Incheon International Airport Corporation, KINTEX, the Institute for Aerospace Industry-Academia Collaboration, and the Seoul Water Reclamation Facilities Corporation.

### Career ▼

Current Position: Professor at Duksung Women's University

#### ■ Key Research Publications:

An Empirical Study on the Development of an Air Logistics System Using Drones

An Analysis of the Impact of Transportation Service Factors on Customer Satisfaction in Drone Delivery

A Study of Location-Based Air Logistics Systems with Light-ID and RFID on Drone Systems: The Case of an Air Cargo Warehouse

A Study on the Application of LED-ID- and RFID-Based Drones for the Efficient Management of Air Cargo Unit Load Devices (ULDs)

#### ■ Book: Air Logistics and Transportation Operations, Doonam Publishing

Samsung SDS Co., Ltd. does not guarantee the reliability or accuracy of the information, data, or facts published in this "i4L White Paper." The company is not responsible for any actions taken by recipients or users based on this information. All intellectual property rights, including copyrights to this document, belong to Samsung SDS Co., Ltd. Please note that using, copying, disclosing, or distributing this document for purposes other than those for which it was provided may constitute copyright infringement against Samsung SDS Co., Ltd.  
Copyright © 2026 Samsung SDS Co., Ltd. All rights reserved.



# Contents

|                                                                                      |    |
|--------------------------------------------------------------------------------------|----|
| 01. China Plus X: Reshaping Asia's Manufacturing Landscape .....                     | 01 |
| 02. The Rise of Asia's Air Cargo Market:<br>A Data-Driven Perspective .....          | 06 |
| 03. Evolving Production Roles and Air Cargo<br>Demand by Country .....               | 11 |
| 04. Multi-Hub Competition and Opportunities for Korea .....                          | 16 |
| 05. Conclusion: A "More Connected Asia" Rather<br>Than a Shift Away from China ..... | 20 |

# 01.

## China Plus X: Reshaping Asia's Manufacturing Landscape

### From Finished-Goods Exporter to Manufacturing Platform: China's Changing Role

Air cargo associated with China accounts for a substantial share of the global air cargo market. In particular, growth in air cargo volumes connecting China with North America and the trans-Pacific region has already been widely reported, driven by shipments of electronics and consumer goods assembled in China and the explosive growth of e-commerce. In recent years, China has evolved beyond being merely a production base offering low manufacturing costs into a vast manufacturing platform where component sourcing, assembly, exports, and logistics are integrated in one place.

China's role as a final assembly and manufacturing hub is changing. Manufacturing investment in Southeast Asian countries is expanding, while India is also emerging as a major new production base. These developments have commonly been described as a "shift away from China" or "China Plus One."

However, a closer examination of actual supply chains reveals a very different picture from one in which factories in China are relocating en masse to other countries. Components and production equipment manufactured in China are being transported to Vietnam and Thailand, while semiconductors and equipment produced in Korea and Taiwan pass through manufacturing facilities in Malaysia before being supplied to factories in China or elsewhere in Asia. Chinese companies are also expanding their own production facilities in Southeast Asia, creating new regional supply chains.

McKinsey & Company notes that although production bases are expanding into Southeast Asia, China remains a critical global manufacturing hub due to its already-established scale of production and manufacturing ecosystem. It also observes that Chinese companies themselves are actively participating in the expansion of manufacturing across Southeast Asia<sup>1</sup>.

Therefore, rather than interpreting recent developments simply as "factories moving from China to other countries," they should be understood as the expansion of the Asian manufacturing network—originally developed around China—into Southeast Asia and India. From an air cargo perspective, the key issue is not whether the volume of finished goods originating from China is rising or falling. What matters is that Asia as a whole, including China, is being reorganized into a vast production, assembly, and consumption network, making intra-Asian air cargo connections increasingly dense. In other words, as production processes are divided across multiple countries, it is necessary to examine the new flows of components and intermediate goods emerging between China and Vietnam, Malaysia, Thailand, Indonesia, and India.

### From China Plus One to China Plus X

The term most commonly used to describe the diversification of companies' production bases has been "China Plus One." This strategy involves maintaining a primary production base in China while establishing an additional production hub in another country, such as Vietnam or India, thereby reducing excessive dependence on China alone.

However, after experiencing a series of disruptions—including global supply chain breakdowns following the COVID-19 pandemic, U.S.–China trade tensions and changes in tariff policies, the Russia–Ukraine war, and transportation disruptions in the Red Sea and the Middle East—companies have adopted increasingly complex responses. Boston Consulting Group (BCG) does not describe recent global sourcing strategies as simply replacing Chinese suppliers with those in other countries. Instead, it suggests that companies can combine various approaches, such as continuing to work with established large-scale suppliers in China while securing a second supplier within China, or using a primary supplier in China alongside an emergency backup supplier outside China<sup>4</sup>.

[Table 1] Sourcing Strategy Options for Enhancing Supply Chain Resilience

| Strategic Option                                                  | Key Description                                                           |
|-------------------------------------------------------------------|---------------------------------------------------------------------------|
| ① Retain a Single Large Supplier in China                         | Maintain the existing primary supply chain                                |
| ② Secure a Second Supplier in China                               | Diversify suppliers within the same country                               |
| ③ Secure a Backup Supplier Outside China                          | Establish a backup supply chain for contingencies                         |
| ④ Use Specialized Suppliers Outside China for Selected Items Only | Selectively diversify sourcing by item                                    |
| ⑤ Develop a Second Large-Scale Supplier Outside China             | Build an alternative supply chain comparable in scale to the one in China |

Source: Boston Consulting Group (2024), Revitalizing Global Sourcing to Balance Cost and Resilience, Exhibit 3

BCG particularly emphasizes that it is difficult for other countries to replicate China's manufacturing base in the short term. Even though production costs in China have risen compared with the past, its well-established infrastructure, industrial clusters, and supplier networks continue to provide significant competitive advantages<sup>4</sup>.

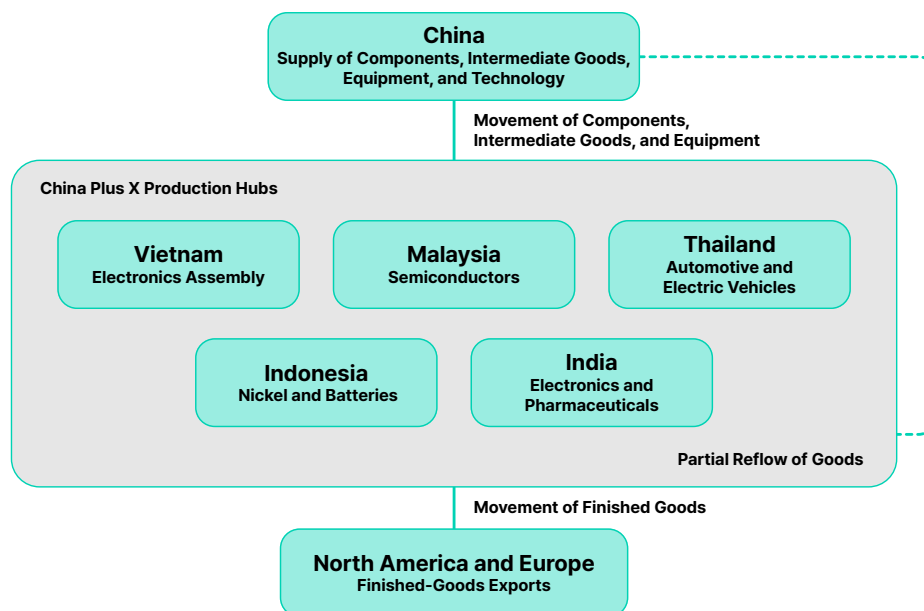
Ultimately, the choice facing most companies is not a binary one of whether to leave China or remain there. Instead, they combine different production locations based on product-specific risks, costs, technology requirements, lead times, and market accessibility.

## 01. China Plus X: Reshaping Asia's Manufacturing Landscape

This paper describes this shift through the concept of “China Plus X.” Here, X does not represent a single country, but rather a combination of multiple production hubs, including Vietnam, Malaysia, Thailand, Indonesia and India. This approach maintains China’s manufacturing base while adding the comparative advantages of individual countries to the production processes where they are most needed.

As BCG’s framework demonstrates, production-base diversification does not involve choosing exclusively between Chinese and non-Chinese supply chains. Because companies can combine multiple approaches depending on the product, the key word in China Plus X may in fact be “Plus.” It does not mean eliminating China; rather, it means connecting additional production locations to China’s existing manufacturing base. Products that remain economical to manufacture in China continue to be produced there, while production processes with higher supply risks or for which local market access is important are added in other countries. Ultimately, China Plus X is less a strategy for finding a “new factory of the world” to replace China than one for combining the distinct production capabilities of multiple countries, including China.

[Figure 1] The Circular Structure of Asian Supply Chains in the China Plus X Era



### China Is Not Disappearing—Its Role Is Changing

Care should also be taken not to interpret the growth of Southeast Asian manufacturing and the decline of Chinese manufacturing as the same phenomenon. According to McKinsey & Company, Chinese companies’ foreign direct investment in Southeast Asian manufacturing reached approximately USD 24 billion in 2023. This represented roughly one-third of China’s total outward foreign direct investment in manufacturing<sup>1</sup>. In other words, Chinese companies are not merely passive players losing their production base as manufacturing activity expands into Southeast Asia. They are also among the region’s major investors, establishing factories and expanding production capabilities there themselves.

The trade structure provides an even clearer picture of the relationship between China and Southeast Asia. According to McKinsey’s analysis, 88% of Southeast Asia’s trade flows remain within the Asia-Pacific region, while approximately 30% of its export flows are directed to China<sup>1</sup>. These figures show that the growth of Southeast Asian manufacturing does not necessarily weaken its economic ties with

## 01. China Plus X: Reshaping Asia's Manufacturing Landscape

---

China. Under this structure, even when a Chinese company establishes a factory in Vietnam or Thailand, machinery, production equipment, molds, electronic components, and intermediate goods manufactured in China may be supplied to the facility. Intermediate or finished goods produced locally may then be shipped back to China or exported to the United States and Europe.

A study by Oh (2025), published in the international academic journal *Asia Europe Journal*, reaches a similar conclusion. During the 2010s, China was the most important supplier of components and intermediate goods to Southeast Asia's manufacturing sector, with more than 75% of goods shipped from China to Southeast Asia consisting of intermediate and capital goods used in local manufacturing. This indicates that trade between China and Southeast Asia depends heavily on interconnected production networks rather than simply on exports of consumer goods<sup>5</sup>.

However, Oh (2025) notes that, through the 2010s, China's role was closer to that of a key supplier than the largest investor controlling Southeast Asian manufacturing. The data show that China was a major supplier of raw materials and components to Southeast Asia at the time, but its role as a "key investor" with substantial decision-making authority over production was more limited than commonly assumed. Japanese, American, Korean, and European companies also accounted for significant shares of manufacturing investment in Southeast Asia. At the same time, the study points out that as investment by Chinese companies in electric vehicles, batteries, solar power, and digital industries has recently increased, China has been expanding its role from a component supplier to an investor directly involved in local production and supply chain operations<sup>5</sup>.

It is therefore unrealistic either to view Southeast Asia as being unilaterally dependent on Chinese manufacturing or to assume that an entirely independent production region is emerging to replace China. In practice, the production activities of companies from China, Korea, Japan, Taiwan, the United States, and Europe are intricately connected at different stages, while China's own role is changing. In the past, China was widely perceived as the "factory of the world," importing components from around the globe, assembling finished goods, and exporting them back to international markets. Today, that role has expanded to include supplying neighboring countries' production facilities with components, intermediate goods, machinery, equipment, and technology manufactured in China, while Chinese companies themselves increasingly participate in local production.

This shift also has important implications for the air cargo market. Even if some finished goods previously exported from China are replaced by products made in Vietnam or Thailand, new demand for intra-Asian air cargo can emerge as greater volumes of electronic components, precision machinery, prototypes, and urgent production materials are transported from China to these countries.

# 02.

## The Rise of Asia's Air Cargo Market: A Data-Driven Perspective

## Asia-Pacific Emerges as the Growth Engine of Global Air Cargo

According to the International Air Transport Association (IATA), global air cargo demand increased by 3.4% year-on-year in 2025, measured in cargo tonne-kilometres (CTK), reaching an all-time high. During the same period, cargo demand among Asia-Pacific airlines grew by 8.4%, the highest growth rate of any region worldwide. These airlines accounted for 35.9% of global air cargo market CTK<sup>2</sup>.

[Table 2] Air Cargo Performance by Region in 2025 (Full-Year Cumulative, Year-on-Year)

(Unit: %)

| Region                          | Global Market Share | Demand Growth (CTK) | Capacity Growth (ACTK) <sup>1)</sup> | Cargo Load Factor (CLF) <sup>2)</sup> |
|---------------------------------|---------------------|---------------------|--------------------------------------|---------------------------------------|
| Total Market                    | 100.0               | 3.4                 | 3.7                                  | 45.7                                  |
| Asia-Pacific                    | 35.9                | 8.4                 | 7.4                                  | 47.6                                  |
| Europe                          | 21.4                | 2.9                 | 3.1                                  | 53.4                                  |
| Middle East                     | 13.2                | 0.3                 | 4.5                                  | 45.1                                  |
| North America                   | 24.5                | -1.3                | -1.1                                 | 40.2                                  |
| Africa                          | 2.1                 | 6.0                 | 7.8                                  | 42.9                                  |
| Latin America and the Caribbean | 2.9                 | 2.3                 | 4.5                                  | 36.0                                  |

Source: IATA (2026a), Global Air Cargo Demand Achieved Record Volume in 2025

1) ACTK(Available Cargo Tonne-Kilometres): The maximum cargo-carrying capacity that an airline can provide.

2) CLF(Cargo Load Factor): The ratio of cargo actually transported (CTK) to available cargo capacity.

The 35.9% figure does not represent the share of cargo originating from or arriving in Asia. Rather, it is calculated based on the region in which each airline is registered, as the International Air Transport Association (IATA) compiles its regional statistics according to airline nationality rather than route origin or destination. Nevertheless, the fact that Asia-Pacific airlines account for more than one-third of global air cargo traffic clearly demonstrates Asia's prominence in the global air cargo market.

Even more notable are the changes by air cargo route. In 2025, air cargo demand on the Europe-Asia route increased by 10.3% year-on-year, while demand on routes within Asia grew by 10.0%. Demand on the Europe-North America route rose by 6.8%, and the Middle East-Asia route recorded growth of 5.8%. By contrast, demand on the Asia-North America route declined by 0.8%<sup>2</sup>.

## 02. The Rise of Asia's Air Cargo Market: A Data-Driven Perspective

[Table 3] Growth Rates and Market Shares of Major International Air Cargo Trade Lanes in 2025

(Unit: %)

| Trade Lane           | 2025 Demand Growth (CTK) | 2025 Global Market Share | Change in Market Share (2024-2025) |
|----------------------|--------------------------|--------------------------|------------------------------------|
| Europe-Asia          | +10.3                    | 21.5                     | +1.1                               |
| Within Asia          | +10.0                    | 7.4                      | +0.4                               |
| Europe-North America | +6.8                     | 13.5                     | +0.3                               |
| Middle East-Asia     | +5.8                     | 7.4                      | +0.1                               |
| Asia-North America   | -0.8                     | 23.4                     | -1.2                               |
| Within Europe        | -1.4                     | 1.9                      | -0.1                               |
| Africa-Asia          | -1.9                     | 1.3                      | -0.1                               |
| Europe-Middle East   | -3.4                     | 5.2                      | -0.4                               |

Source: IATA (2026a), Global Air Cargo Demand Achieved Record Volume in 2025, Trade Lane Growth

The International Air Transport Association (IATA) assessed these developments as a clear shift in global air cargo flows in 2025. While challenges arising from trade relations with the United States—including higher tariffs, the reduction of the de minimis exemption, and continued policy uncertainty—affected the established Asia-North America trade lane, intra-Asian and Europe-Asia air cargo grew relatively rapidly.

Air cargo flows have begun changing rapidly again in 2026. According to data released by IATA on July 29, 2026, global air cargo demand increased by 8.5% year-on-year in June, while cargo demand among Asia-Pacific airlines rose by 7.9%<sup>3</sup>.

[Table 4] Air Cargo Performance by Region in June 2026 (Year-on-Year)

(Unit: %)

| Region        | Global Market Share | Demand Growth (CTK) | Capacity Growth (ACTK) | Cargo Load Factor (CLF) |
|---------------|---------------------|---------------------|------------------------|-------------------------|
| Total Market  | 100.0               | 8.5                 | 4.4                    | 46.9                    |
| Asia-Pacific  | 35.8                | 7.9                 | 4.3                    | 51.8                    |
| Europe        | 21.4                | 6.9                 | 3.7                    | 50.5                    |
| Middle East   | 13.2                | 5.6                 | 2.5                    | 46.5                    |
| North America | 24.6                | 13.1                | 6.2                    | 40.7                    |

Source: IATA (2026b), Air Cargo Demand Strengthens in June, Up 8.5%

In particular, the Asia-North America route grew by 14.7%, recording the highest growth rate among major international air cargo routes. Intra-Asian routes increased by 7.2%, while the Europe-Asia route grew by 7.1%. Intra-Asian routes extended their growth streak to 32 consecutive months, while the Europe-Asia route recorded growth for the 40th consecutive month<sup>3</sup>.

## 02. The Rise of Asia's Air Cargo Market: A Data-Driven Perspective

[Table 5] Growth Rates and Market Shares of Major Air Cargo Trade Lanes in June 2026

(Unit: %)

| Trade Lane           | Year-on-Year Change | Remarks                          | Global Market Share |
|----------------------|---------------------|----------------------------------|---------------------|
| Asia-North America   | +14.7               | Growth for 5 consecutive months  | 23.5                |
| Europe-Asia          | +7.1                | Growth for 40 consecutive months | 21.5                |
| Within Asia          | +7.2                | Growth for 32 consecutive months | 7.3                 |
| Middle East-Asia     | -4.1                | Decline for 4 consecutive months | 7.4                 |
| Africa-Asia          | +0.9                | Growth for 12 consecutive months | 1.3                 |
| Europe-North America | 0.0                 | -                                | 13.5                |
| Europe-Middle East   | -41.1               | Decline for 4 consecutive months | 5.2                 |

Source: IATA (2026b), Air Cargo Demand Strengthens in June, Up 8.5%. Global market shares are based on full-year 2025 CTK.

※ As IATA subsequently published revised full-year 2025 data, figures in the January release (Table 3) may differ slightly from those in the July release.

The fact that Asia-North America air cargo, which declined in 2025, rebounded rapidly in 2026 offers several important implications. Rather than one route disappearing and another taking its place, air cargo flows are being rapidly realigned in response to tariffs, e-commerce, semiconductors and advanced technology products, and geopolitical risks. From an expert perspective, this volatility itself can be regarded as a defining characteristic of the China Plus X structure. In other words, the center of gravity in global air cargo is shifting beyond the conventional structure of shipments moving from a single country to North America or Europe toward a more densely connected network of production hubs across Asia.

One notable exception is the routes passing through the Middle East. The Europe-Middle East route plunged by 41.1%, contracting for the fourth consecutive month, while the Middle East-Asia route also declined by 4.1%, following the same trend. This was the result of geopolitical conflicts in the Middle East worsening transportation conditions along these routes, in contrast to the robust growth of intra-Asian and Europe-Asia routes. The diversification of air cargo networks across multiple axes in the China Plus X era also serves as a form of buffer, preventing disruptions along any single axis from undermining overall air cargo growth in Asia.

### Beyond E-Commerce: The Shift Toward High-Value, Advanced Technology Cargo

Over the past several years, e-commerce has been one of the key drivers of growth in Asia's air cargo market. In particular, the expansion of e-commerce models that deliver small, lightweight consumer goods manufactured in China directly to consumers in the United States and Europe generated substantial air cargo demand. Since 2025, however, changes in U.S. tariff policy and the de minimis exemption for low-value shipments have begun to reshape the e-commerce-driven air cargo structure. This does not mean that air cargo demand in Asia itself is weakening.

In June 2026, global merchandise trade increased by 5.2% year-on-year, while air cargo CTK grew by 8.5%. The International Air Transport Association (IATA) explained that recent growth in air cargo demand has been driven by high-value technology products and urgent shipments<sup>3</sup>.

This also means that future assessments of Asia's air cargo market must consider not only cargo volumes but also the types of goods being transported. High-value, time-sensitive cargo—including semiconductors and electronic components, equipment required for data centers and advanced technology industries, and biopharmaceuticals—depends more heavily on air transportation than general consumer goods. Because the production processes for these industries are distributed across multiple countries, they are likely to further strengthen the relationship between China Plus X and air cargo.

Even if growth in small e-commerce parcels slows, technology products and industrial goods generated by supply chain restructuring could fill much of the gap. The air cargo industry therefore needs to redesign its cargo portfolio accordingly.

# 03.

## Evolving Production Roles and Air Cargo Demand by Country



## Industry-Specific Positions of Vietnam, Malaysia, Thailand, Indonesia, and India

In Asia's emerging manufacturing landscape, not every country is competing with China in the same industries. As each country differs in its industrial base, technological capabilities, workforce, natural resources, and domestic market, its role within the production network is also evolving differently.

[Figure 2] Geographic Distribution and Key Industries of China Plus X Production Hubs



### ★ [Vietnam] Electronics and Assembly Manufacturing

Vietnam has been one of the primary beneficiaries of the China Plus X strategy. Its proximity to southern China has helped drive the rapid expansion of its industrial base for electronics, mobile phones, and consumer goods. A production structure is emerging in which components and manufacturing equipment are supplied from countries such as China and Korea, while products assembled in Vietnam are subsequently shipped to the United States, Europe, China, and other markets.

As Oh (2025) notes, intermediate goods and capital goods supplied by China have played an increasingly important role in the growth of Vietnam's electronics industry. From an air cargo perspective, two flows are becoming equally important: electronic components and urgent production supplies moving between southern China and northern Vietnam, and finished products shipped from Hanoi and Ho Chi Minh City to North America and Europe.



### **[Malaysia] A Key Link in the Semiconductor Supply Chain**

Malaysia has long established a strong foundation in the electronics industry and semiconductor assembly and testing. Semiconductors are particularly closely linked to air cargo because of their high value and the importance of precise time management between production stages.

Materials and equipment production facilities in Korea, Taiwan, Japan, and China, semiconductor manufacturing processes in Malaysia, and end markets in China, the United States, and Europe can all be connected within a single supply chain. As production becomes more dispersed across multiple countries, the movement of wafers, precision components, manufacturing equipment, and urgently needed maintenance parts is also likely to increase.



### **[Thailand] From Automobiles to Electric Vehicles**

Thailand has traditionally developed into one of Southeast Asia's leading automotive manufacturing hubs, driven largely by Japanese automakers. More recently, Chinese electric vehicle manufacturers have expanded their investments across Southeast Asia, including Thailand, adding new production networks for electric vehicles, batteries, and automotive electronics to the existing internal combustion engine vehicle supply chain<sup>1</sup>.

Because finished vehicles are primarily transported by sea, increased automobile production does not directly translate into higher air cargo volumes. However, electronic control units, prototypes, urgently needed components, manufacturing equipment, and after-sales service parts required to operate production lines are closely linked to air transport. In the automotive industry, where production line disruptions can be extremely costly, even a difference of a few hours in transit time can be highly significant.



### **[Indonesia] From Nickel to the Battery Supply Chain**

Leveraging its abundant nickel resources, Indonesia is expanding its influence in the upstream segment of the electric vehicle battery supply chain. As nickel ore is typically transported by sea, raw material production itself does not generate significant air cargo volumes.

However, as the number of smelting facilities and battery material production plants increases, demand may also grow for transporting the precision machinery, electronic equipment, samples, and urgently needed maintenance parts required to construct and operate these facilities. As a resource-based industry evolves into advanced manufacturing, the volume of high-value cargo suitable for air transport also increases.



### **[India] An Emerging Production Hub and a Vast Consumer Market**

India differs somewhat from Southeast Asian countries, as it combines a vast domestic market, a large pool of IT and engineering talent, and established electronics and pharmaceutical manufacturing capabilities. For companies, India is significant not only as a production base outside China but also as a manufacturing hub that provides direct access to its domestic market.

ACI Asia-Pacific & Middle East forecasts that South Asian air cargo will grow at a compound annual growth rate of 5.5% through 2028—the highest among Asia-Pacific subregions. Southeast Asia is projected to grow by 5.2% annually, while China is expected to grow by 4.4%<sup>6</sup>.

### 03. Evolving Production Roles and Air Cargo Demand by Country

[Table 6] Key Production Regions and Air Cargo in the China Plus X Era

| Region    | Primary Production Role                                   | Key Air Cargo-Related Goods                                                     |
|-----------|-----------------------------------------------------------|---------------------------------------------------------------------------------|
| China     | Components, intermediate goods, equipment, and technology | Electronic components, manufacturing equipment, and high-value products         |
| Vietnam   | Assembly of electronics and consumer goods                | Components, production supplies, and finished products                          |
| Malaysia  | Semiconductors and electronics                            | Semiconductors, precision components, and manufacturing equipment               |
| Thailand  | Automobiles and electric vehicles                         | Automotive electronics, prototypes, urgent parts, and after-sales service parts |
| Indonesia | Nickel and battery materials                              | Precision equipment, samples, and maintenance parts                             |
| India     | Electronics, pharmaceuticals, and the domestic market     | Electronics, pharmaceuticals, and high-value cargo                              |

Source: McKinsey & Company (2024); Oh (2025); ACI APAC & MID (2025)

This production landscape makes the meaning of China Plus X clearer. Vietnam is not replacing China's entire manufacturing capacity, nor is Malaysia or India becoming the new single "factory of the world." Instead, each country is expanding its role in specific industries or stages of production. As a result, the production structure in which multiple countries must be connected to create a single finished product is becoming increasingly pronounced.

## New Air Cargo Connections Created by the Dispersion of Production

It is easy to assume that when production facilities are dispersed across multiple countries, supply chains will naturally become shorter. However, the opposite can occur within global value chains (GVCs), as components and intermediate goods may repeatedly cross national borders during the production of a single finished product. For example, electronic components manufactured in China may be shipped to Vietnam for assembly, while semiconductors from Korea or Taiwan may undergo back-end processing in Malaysia before being transported to finished-product manufacturing facilities in China or Vietnam. Automotive electronics and manufacturing equipment produced in China may be shipped to electric vehicle plants in Thailand, while materials produced in Indonesia may be linked to battery manufacturing facilities in China and elsewhere in East Asia. Oh's (2025) finding that intermediate and capital goods accounted for more than 75% of China's exports to Southeast Asia clearly illustrates this structure.

Of course, dispersing production processes across multiple countries does not mean that all additional cargo volumes will shift to air transport. Heavy, low-value goods such as minerals, steel, and chemical raw materials are primarily transported by sea, while ocean shipping is also far more economical for the large-scale transportation of price-sensitive consumer goods. Semiconductors, electronic components, pharmaceuticals, prototypes, and urgently needed parts required to prevent production line stoppages are different. The value of these goods is significantly higher than their transportation cost, and the losses caused by a day—or even a few hours—of halted production may exceed the cost of airfreight. Therefore, as production hubs become more geographically dispersed and the distances between production stages increase, demand for air transport of such time-sensitive cargo is likely to grow.

### 03. Evolving Production Roles and Air Cargo Demand by Country

---

The influx of manufacturing investment into Southeast Asia also supports this structural transformation. According to the ASEAN Investment Report 2025, jointly published by the Association of Southeast Asian Nations (ASEAN) and the United Nations Conference on Trade and Development (UNCTAD), foreign direct investment (FDI) inflows into ASEAN increased by 8% to USD 226 billion in 2024. Manufacturing FDI, in particular, rose by approximately 150% to USD 44 billion. The report assesses that ASEAN is emerging as a major hub in global supply chains, particularly in supply chain-intensive industries such as semiconductors, automobiles, and apparel<sup>7</sup>.

Increased manufacturing investment does not simply mean more factories. It also creates demand for component suppliers, manufacturing equipment, warehouses and logistics centers, as well as customs clearance and transportation services. Logistics is both an industry that follows the relocation of production hubs and a prerequisite for new production networks to function. This is precisely what China Plus X means for the air cargo market. Rather than Chinese routes disappearing and being replaced by routes to Vietnam or India, new connections linking China and Southeast Asia, Southeast Asian countries with one another, and Southeast Asia with North America and Europe are being added to the existing Asian air cargo map.

# 04.

## Multi-Hub Competition and Opportunities for Korea



## The Changing Role of Chinese Hubs and Multi-Hub Networks

As Asian supply chains expand across multiple countries and industrial cities, the role and competitiveness of airports will also evolve. In the past, annual cargo throughput and the number of routes served by freighter aircraft were the primary measures of a cargo airport's competitiveness. Although cargo-handling volume itself will remain important, it may no longer be sufficient on its own.

Facilities capable of safely handling semiconductors and electronic components, cold-chain infrastructure for pharmaceuticals and perishables, automated systems for the rapid sorting of e-commerce shipments, and digital platforms that connect cargo information among customs authorities, airlines, and freight forwarders in real time will become increasingly important.

[Table 7] Air Cargo Growth Outlook by Asia-Pacific Subregion (2025–2028 Annual Average, %)

| Subregion      | Average Annual Growth Rate(%) |
|----------------|-------------------------------|
| South Asia     | 5.5                           |
| Southeast Asia | 5.2                           |
| China          | 4.4                           |
| Oceania        | 3.7                           |
| East Asia      | 2.6                           |

Source: ACI APAC & MID (2025), Asia-Pacific and Middle East Will Continue to Lead Global Air Cargo Growth through 2028

A notable point in the forecast by Airports Council International (ACI) is that, despite Southeast Asia's strong growth, China's air cargo market is also projected to expand at an average annual rate of 4.4% through 2028. This suggests that the rise of Southeast Asia will not necessarily lead directly to a weakening of China's air cargo market or its role as a logistics hub.

China's major cargo hubs—including Shanghai, Shenzhen, Guangzhou, Zhengzhou, and Hong Kong—are supported not only by large-scale manufacturing bases but also by a vast domestic market. Therefore, even if some production processes expand into Vietnam, Malaysia, Thailand, and other countries, logistics flows between China and these production hubs may actually increase. New cargo flows will emerge as key components and manufacturing equipment produced in China are shipped to factories in Southeast Asia, while intermediate goods and finished products manufactured there are transported back to China or onward to global markets such as North America and Europe.

Ultimately, the future transformation of Asian air logistics cannot be explained as a simple substitution in which Chinese hubs decline and Southeast Asian airports take their place. As airports in Hanoi, Ho Chi Minh City, Penang, Bangkok, Jakarta, and India's major industrial cities grow alongside new production hubs, connections between these emerging airports and established major hubs such as Shanghai, Hong Kong, and Shenzhen are also likely to become more active. Accordingly, Asia's air cargo network is expected to gradually evolve beyond a structure centered on a small number of major hub airports toward a multi-hub network in which established hubs are interconnected with airports serving multiple production centers.

### Core Capabilities Required for Korean Air Logistics

These changes also carry important implications for Korea's airlines, airports, freight forwarders, and logistics companies. First, China Plus X should not be viewed solely in terms of declining Chinese cargo volumes. Even if some of China's final assembly functions expand into Vietnam, Malaysia, Thailand, Indonesia, India and other countries, the volume of components, intermediate goods, and machinery shipped from China to these production regions may actually increase.

Korea also needs to approach China and Southeast Asia as an interconnected production region rather than treating them solely as separate markets. Instead of viewing China-Korea, Korea-Southeast Asia, and China-Southeast Asia air routes as independent markets, logistics services capable of managing production facilities and component inventories dispersed across multiple countries as a single network will become increasingly important.

The role of Incheon International Airport also needs to be reconsidered. In 2024, transfer cargo accounted for 1.18 million tonnes—or approximately 40%—of the airport's 2.95 million tonnes of international air cargo. This transfer ratio, higher than those of neighboring competing airports, has reinforced Incheon's position as a logistics hub in Northeast Asia<sup>8</sup>. Beyond its traditional role as an intermediate transfer point for China-North America cargo, Incheon can expand its role as a Northeast Asian logistics hub connecting Korea's semiconductor, battery, and automotive parts industries; China's manufacturing and component supply chains; and production hubs in Vietnam, Malaysia, Thailand, Indonesia, India and other countries—then linking them onward to markets in North America and Europe.

Transportation modes must also be combined more flexibly. In addition to long-haul air transport using large freighter aircraft, it is necessary to consider passenger aircraft belly cargo, small and medium-sized freighters, trucking and bonded transportation, and sea-air multimodal transport. In 2025, dedicated freighter cargo volume at Incheon Airport fell by 0.7% year on year. However, increased belly cargo resulting from expanded passenger operations offset the decline, lifting total cargo throughput by 0.3% to 2,954,684 tonnes<sup>9</sup>. This demonstrates that a strategy combining multiple transportation modes rather than relying on a single mode can serve as a buffer against fluctuations in cargo volumes. From a shipper's perspective, there is no reason to transport every shipment by the fastest air service, nor can every product be shipped solely by the lowest-cost ocean service. The most appropriate combination of transportation modes must be selected according to product value, inventory levels, the risk of production disruption, transit time, and delivery deadlines.

In the China Plus X era, it is no longer sufficient to identify only the country from which a shipment ultimately departs. Companies must also be able to determine where components manufactured in China were processed, how much value was added at each stage of production, and how rules of origin and tariffs apply. Ultimately, the competitiveness of logistics companies will no longer be determined solely by their ability to transport cargo quickly.

### Core Capabilities Required for Korean Logistics Companies

- **Supply Chain Visibility** — The ability to manage production facilities and inventories dispersed across multiple countries in real time as a single integrated network.
- **Customs and Rules-of-Origin Data Management** — The ability to accurately track value added across multi-stage production processes, rules of origin, and applicable tariffs.
- **Inventory Optimization and Risk Forecasting** — The ability to identify risks associated with specific routes and production hubs in advance and adjust inventory allocation accordingly.
- **Rapid Provision of Alternative Transportation Routes** — The ability to immediately switch among air, ocean, and multimodal transportation when supply chain disruptions occur.

If Korea can provide specialized logistics services for its key industries—including semiconductors, batteries, biopharmaceuticals, automotive parts, and precision machinery—to production hubs in both China and Southeast Asia, it could become a vital link in Asia’s emerging production network.

# 05.

## Conclusion: A “More Connected Asia” Rather Than a Shift Away from China



## 05. Conclusion: A “More Connected Asia” Rather Than a Shift Away from China

---

If the Asian logistics landscape of the China Plus X era could be summed up in a single phrase, it would be “the simultaneous progression of dispersion and connectivity.” Production is expanding from a structure once concentrated primarily in China to multiple locations, including Vietnam, Malaysia, Thailand, Indonesia and India. However, the dispersion of production does not mean weaker connections between China and other Asian countries. On the contrary, as production processes are divided among multiple countries, components, intermediate goods, manufacturing equipment, and finished products cross borders more frequently than before.

In the global sourcing strategies presented by BCG, companies are not withdrawing from Chinese supply chains all at once. Instead, they are maintaining their existing supply bases in China while adding new sources for specific product categories<sup>4</sup>. McKinsey’s analysis also shows that Chinese companies are expanding direct manufacturing investment in Southeast Asia and that production networks between China and Southeast Asia remain closely interconnected<sup>1</sup>. Academic research likewise demonstrates that a significant share of goods moving from China to Southeast Asia consists not of finished consumer products but of intermediate and capital goods used by local manufacturers<sup>5</sup>.

Therefore, the question we should ask in the China Plus X era may not be, “Which country will replace China?” The more important question is who can most efficiently connect China’s manufacturing base, Vietnam’s assembly industry, Malaysia’s semiconductor sector, Thailand’s automotive industry, Indonesia’s battery materials industry, and India’s emerging production base and consumer market.

In the past, the Asian air cargo map was dominated by major routes extending from China’s vast manufacturing base to the United States and Europe. Today, more connections are being added among China, ASEAN member states, Korea, Japan, Taiwan, and India. China is not disappearing from the Asian supply chain. In addition to its traditional role as the “factory of the world,” where products were completed and exported to global markets, China is expanding its role as a platform for an Asian manufacturing network that connects components, intermediate goods, manufacturing equipment, and multiple production hubs.

As the production network developed around China expands into Southeast Asia and India, the routes along which air cargo must move are also becoming far more diverse. Amid these changes, the winning airlines, airports, and logistics companies will not necessarily be those handling the largest cargo volumes.

Companies capable of connecting multiple production hubs quickly and reliably, managing shippers’ inventory as well as customs and rules-of-origin data, and immediately providing alternative transportation routes when supply chain disruptions occur will be well positioned to capture new opportunities. This will be the new air cargo map of “Emerging Asia” that lies ahead.

- [1] McKinsey & Company (2024) Diversifying Global Supply Chains: Opportunities in Southeast Asia, 5 September.
- [2] International Air Transport Association (IATA) (2026a) 'Global Air Cargo Demand Achieved Record Volume in 2025', 29 January. Geneva: IATA. Available at: <https://www.iata.org/en/pressroom/2026-releases/2026-01-29-01/> (Accessed: 9 August 2026).
- [3] International Air Transport Association (IATA) (2026b) 'Air Cargo Demand Strengthens in June, Up 8.5%', 29 July. Geneva: IATA. Available at: <https://www.iata.org/en/pressroom/2026-releases/07-29-air-cargo-demand-strengthens-june/> (Accessed: 9 August 2026).
- [4] Boston Consulting Group (BCG) (2024) Revitalizing Global Sourcing to Balance Cost and Resilience, 13 February. Boston: Boston Consulting Group.
- [5] Oh, Y.A. (2025) 'Reassessing China's economic power in Southeast Asia during the 2010s: insights from the nexus of FDI-driven manufacturing and GVC trade', *Asia Europe Journal*, 23, pp. 349-367. doi:10.1007/s10308-025-00735-2.
- [6] Airports Council International Asia-Pacific & Middle East (ACI APAC & MID) (2025) 'Asia-Pacific and Middle East Will Continue to Lead Global Air Cargo Growth through 2028', 11 December. Available at: <https://aci-asiapac.aero> (Accessed: 9 August 2026).
- [7] United Nations Conference on Trade and Development (UNCTAD) (2025) ASEAN Investment Report 2025: Foreign Direct Investment and Supply Chain Development. Geneva and Jakarta: UNCTAD and ASEAN Secretariat.
- [8] Incheon International Airport (2025) 'About Incheon International Airport: The World's Third-Largest Airport by Cargo Throughput', Available at: [https://www.airport.kr/co\\_ko/719/subview.do](https://www.airport.kr/co_ko/719/subview.do) (Accessed: 26 August 2026).
- [9] Incheon International Airport (2026) '2025 Incheon International Airport Air Transport Performance' Press release, 26 January. Available at: [https://www.airport.kr/co\\_ko/664/subview.do](https://www.airport.kr/co_ko/664/subview.do) (Accessed: 26 August 2026).



 [www.cello-square.com](http://www.cello-square.com)

 [cellosquare@samsung.com](mailto:cellosquare@samsung.com)

 [youtube.com/samsungsds](https://youtube.com/samsungsds)



## Digital Logistics Service Cello Square

We offer differentiated digital logistics services by leveraging Samsung SDS's IT expertise and its top-20 global logistics operation capabilities.

[www.cello-square.com](http://www.cello-square.com)